

Quiz 15
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1) $\lambda = 2$ and $\alpha = 6$

a) possibilities,

6	1, 1, 1, 3
1, 5	2, 2, 2
2, 4	1, 1, 2, 2
1, 1, 4	1, 1, 1, 1, 2
3, 3	1, 1, 1, 1, 1, 1
1, 2, 3	

choose 3, 3

$$J_3(2) = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix} \rightarrow J = \begin{bmatrix} J_3(2) & 0_{3 \times 3} \\ 0_{3 \times 3} & J_3(2) \end{bmatrix}$$

$$J = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$A = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$b) \quad A = PJP^{-1}$$

$$P = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P^{-1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ -1 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$A = PJP^{-1} = \begin{bmatrix} 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 2 & 1 \\ -2 & -1 & 0 & 0 & 0 & 2 \end{bmatrix}$$

$$A - 2I = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ -2 & -1 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$(A - 2I)^2 = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Kernel: $x_3 = 0$, $x_6 = 0$, $-x_3 = 0$

Basis: $[1, 0, 0, 0, 0, 0]^T$, $[0, 1, 0, 0, 0, 0]^T$
 $[0, 0, 0, 1, 0, 0]^T$, $[0, 0, 0, 0, 1, 0]^T$

$$(A - 2I)^3 = 0$$

$$c) A^3 = P K P^{-1}$$

$$A^3 = (P J P^{-1})^3 = P J^3 P^{-1}$$

$$\therefore K = J^3 \text{ For } J = \begin{bmatrix} J_3 & 0 \\ 0 & J_3 \end{bmatrix} \text{ where } J_3 = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$$

$$J_3 = 2I + N, \quad N = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

$$N^2 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$N^3 = \underline{0}$$

$$\begin{aligned} (J_3)^3 &= (2I + N)^3 = 8I + 12N + 6N^2 \\ &= \begin{bmatrix} 8 & 12 & 6 \\ 0 & 8 & 12 \\ 0 & 0 & 8 \end{bmatrix} \end{aligned}$$

$$K = \begin{bmatrix} 8 & 12 & 6 & 0 & 0 & 0 \\ 0 & 8 & 12 & 0 & 0 & 0 \\ 0 & 0 & 8 & 0 & 0 & 0 \\ 0 & 0 & 0 & 8 & 12 & 6 \\ 0 & 0 & 0 & 0 & 8 & 12 \\ 0 & 0 & 0 & 0 & 0 & 8 \end{bmatrix} //$$

